

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start for free" section and create an account.

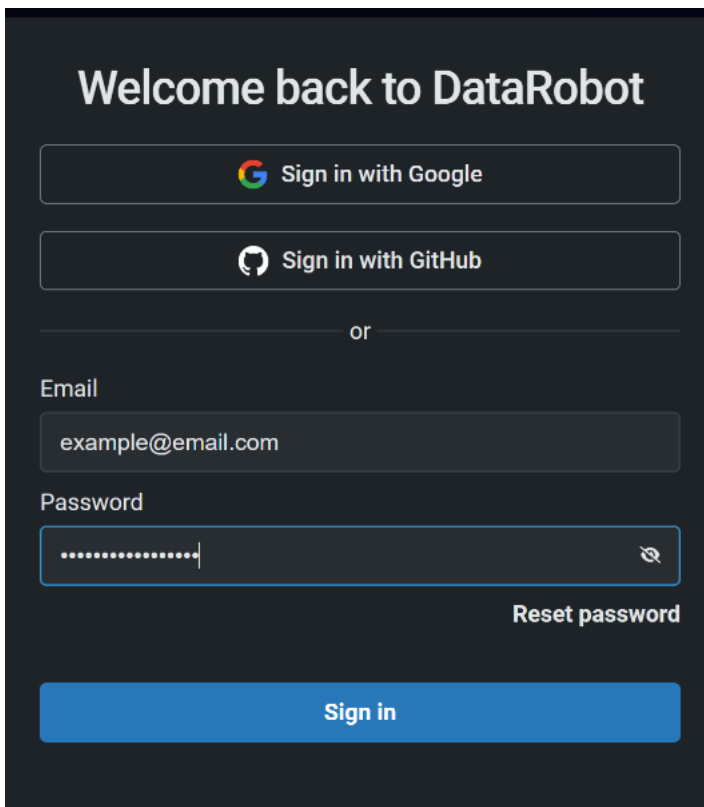
The screenshot shows the DataRobot website's homepage. The header includes the DataRobot logo, navigation links (Platform, Solutions, Customers, Partners, Resources, Company), and links for Support, Log In, Contact Us, and Book a Demo. The main content area features the headline "Experience the DataRobot AI Platform" and a sub-headline "Less Friction, More AI. Get Started Today With a Free 30-Day Trial." Below this, a paragraph states: "Unleash your skills and embark on a new era of AI with a single platform that streamlines your predictive and generative AI workflows." A section titled "Start your free 30 day trial to:" lists four bullet points with checkmarks: "Experience how to fast-track preparing data, running experiments, and testing your models", "See how you can finally automate all your AI Experimentation and AI Production processes in a single solution", "Build predictive and generative AI use-cases all in a single platform", and "Learn how to accelerate your AI plans with hands-on labs, use-case code snippets, and an AI expert community". On the right side, there is a "Start For Free" form with input fields for First Name *, Last Name *, Business Email *, Phone, Company *, Job Title *, and Country *. At the bottom of the form, there is a checkbox labeled "Yes! Please email me news and".

Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 4: A new window will open; select the relevant option for signing up.



Welcome back to DataRobot

Sign in with Google

Sign in with GitHub

or

Email

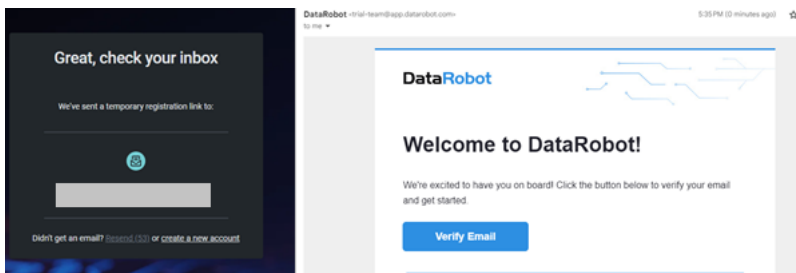
example@email.com

Password

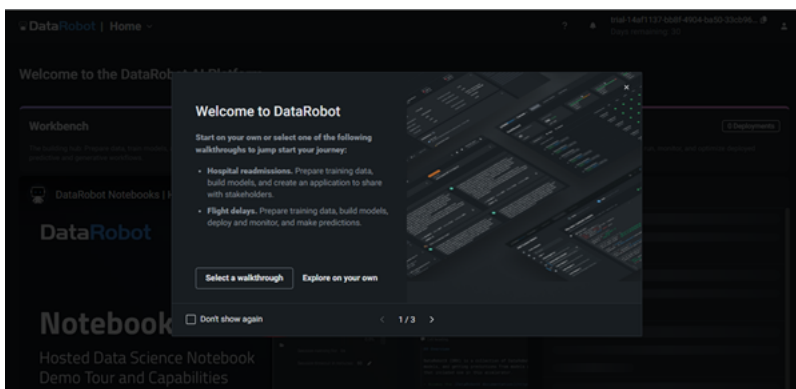
Reset password

Sign in

Step 5: Confirm your email by clicking **Verify Email** in your inbox.

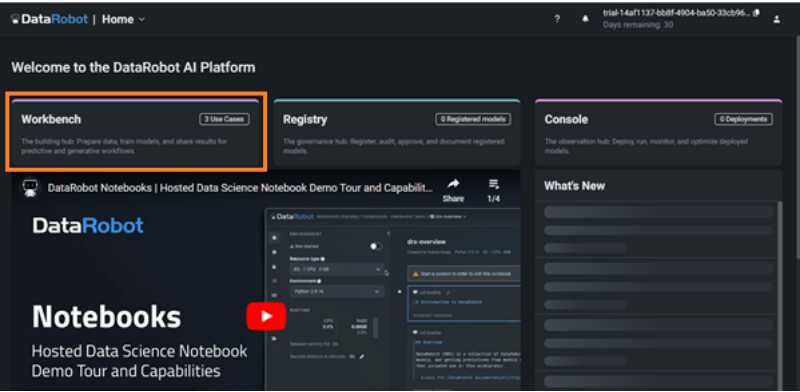


Step 6: Sign up and start your first experience of using the Generative AI tool.
The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

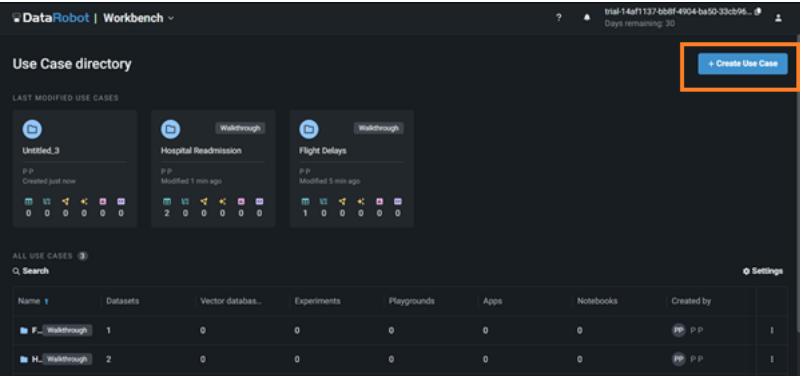


Task 2: Add a data set

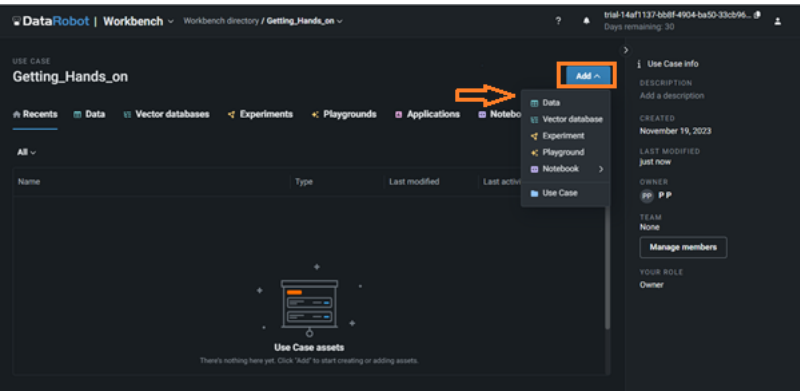
Step 7: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.



Step 8: Click Create Use Case.



Step 9: Click Add and Data to include the data set in your use case.



Step 10: Upload your data set or Connect to the data source; however, for this lab, you can select an in-built sample data set *HOSPITAL_READMISSION_DATA*.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81

Days remaining

Add data

Close

DATA STORES

↑ Upload

+ Connect

Q Search

Data Registry

DataRobot

Demo Connection

Snowflake

Data Registry

Q Search

☐	Name	Created on ↑	Type	Last modified
☐	HOSPITAL_READMISSION_DATA	1d ago	Snapshot	1d ago

< Previous

1

Next >

1 per page ▾

1-1 of 1 items

Step 11: Once you select the data set, you can see a preview of it. You can also view the data set's features, as shown below. Click **Add to Use Case**.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81

Days remaining

Add data

DATA STORES

↑ Upload

+ Connect

Q Search

Data Registry

DataRobot

Demo Connection

Snowflake

HOSPITAL_READMISSION_DATA

Data Registry / HOSPITAL_READMISSION_...

≡ Features

👁 Data preview

race	gender	age	weight	admission_type
Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Male	[50-60)		Emergency
Caucasian	Female	[70-80)		Urgent
Caucasian	Male	[60-70)		Elective
Caucasian	Male	[60-70)		
Caucasian	Female	[40-50)		Not Available
Caucasian	Male	[90-100)		Emergency
Caucasian	Male	[70-80)		Urgent
Caucasian	Female	[80-90)		Urgent

👁 Snapshot sample

51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

about:blank

4/10

DataRobot | Workbench Workbench directory / Untitled_4

< Use Case directory

USE CASE
Untitled_4

Add Data

All Data 1 Vector databases Experiments Playgrounds Applications

Type Search Settings

Name	Created By	Last Modified	Type	Source	Rows
HOSPITAL_READMISSION_DATA	PV Pratiksha V.	now	Snapshot	Snowflake	10000

< Previous 1 Next > 1 per page 1-1 of 1 items

Step 13: Explore the **All Features** menu to display specific features.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA

HOSPITAL_READMISSION_DATA Jul 29th, 2024 10:39 AM Snapshot Data actions

Data preview **Features** Feature lists

Show insights Show features from: All Features + Create feature list

Search

DATAROBOT FEATURE LISTS

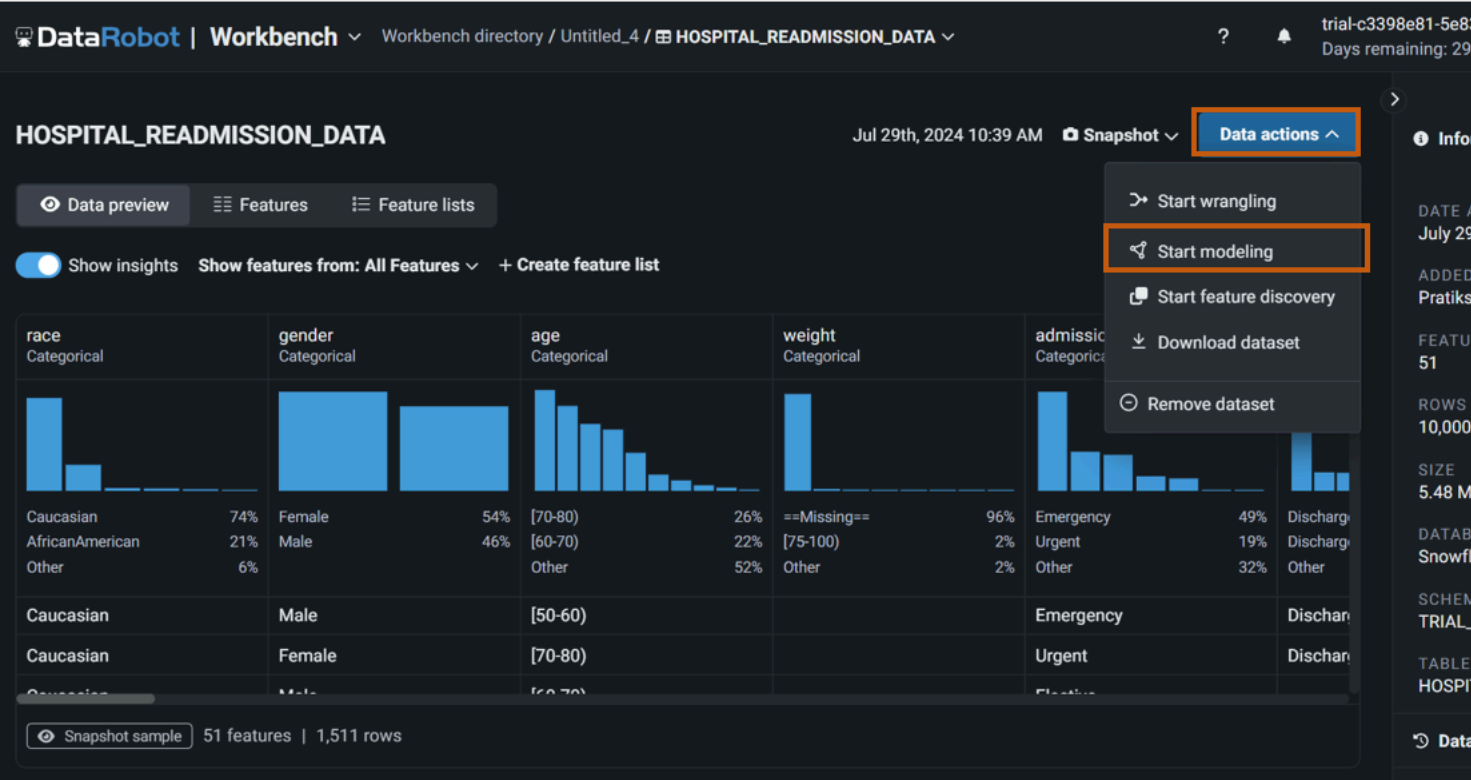
Feature	Count
All Features	51
Informative Features	40
Raw Features	51

race	age	weight	admission_type_id	discharge
Caucasian	[70-80]	==Missing==	Emergency	Discharge
AfricanAmerican	[60-70]	[75-100]	Urgent	Discharge
Other	Other	Other	Other	Other

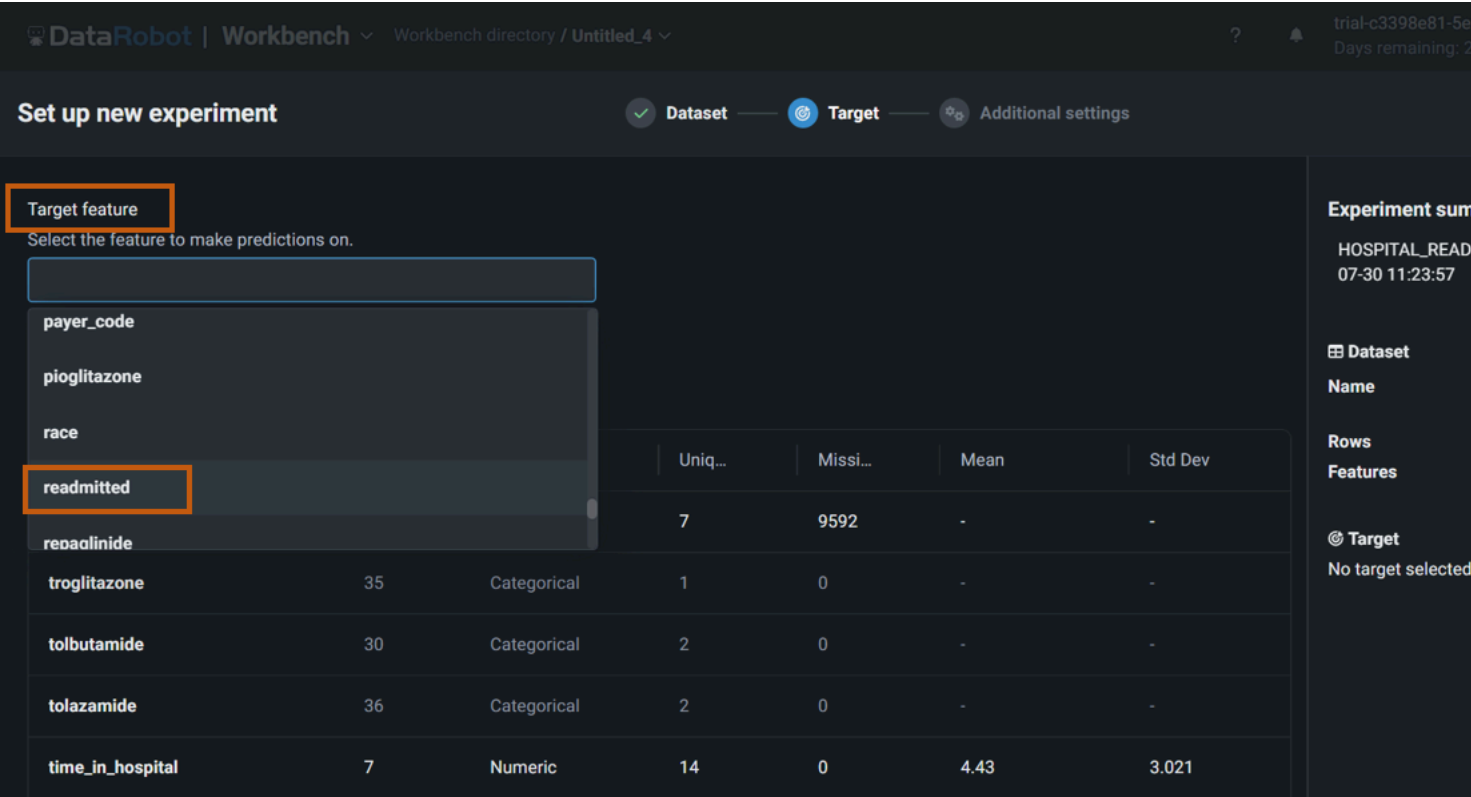
Snapshot sample 51 features | 1,511 rows

Task 3: Work on Data Modeling

Step 14: Click **Start**. You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building. Click **Start** and select **Modelling**. It will take a while to prepare a data set for modelling.



Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.



Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e Days remaining: 2

Set up new experiment

Dataset Target Additional settings

Target feature
Select the feature to make predictions on.

readmitted

Target type: Binary classification

Positive class: ☐ 0 ☒ 1

Modeling mode
Set the mode used for selecting which blueprints to build when training models.

Quick Autopilot

Optimization metric
Set the metric used when training models to evaluate and optimize accuracy.

LogLoss (Accuracy) Recommended

Number of rows

Values of readmitted	Number of rows
False	~6000
True	~4000

Experiment summary
HOSPITAL_READM
07-30 11:23:57

Dataset
Name

Rows
Features

Target
Feature
Target type
Positive class
Modeling mode
Optimization metric
Training feature list

Partitioning

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e Days remaining: 2

Set up new experiment

Dataset Target Additional settings Exit

Data partitioning **Time series modeling** Preview **Additional settings**

Partitioning method
Select the method for assigning rows to partitions when training models.

Stratified sampling
Rows are assigned to ensure similar target distribution across each partition.

Validation type

☒ **Cross-validation**
Trains models on a specified number of folds, maximizing data use but also increasing run time.

☐ **Training-validation-holdout**
Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds
Enter a value from 2 - 50.

Holdout percentage
Set the subset of data that is unavailable during training and validation. Enter a value

Experiment summary
HOSPITAL_READM
07-30 11:23:57

Dataset
Name

Rows
Features

Target
Feature
Target type
Positive class
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Training feature list

Partitioning

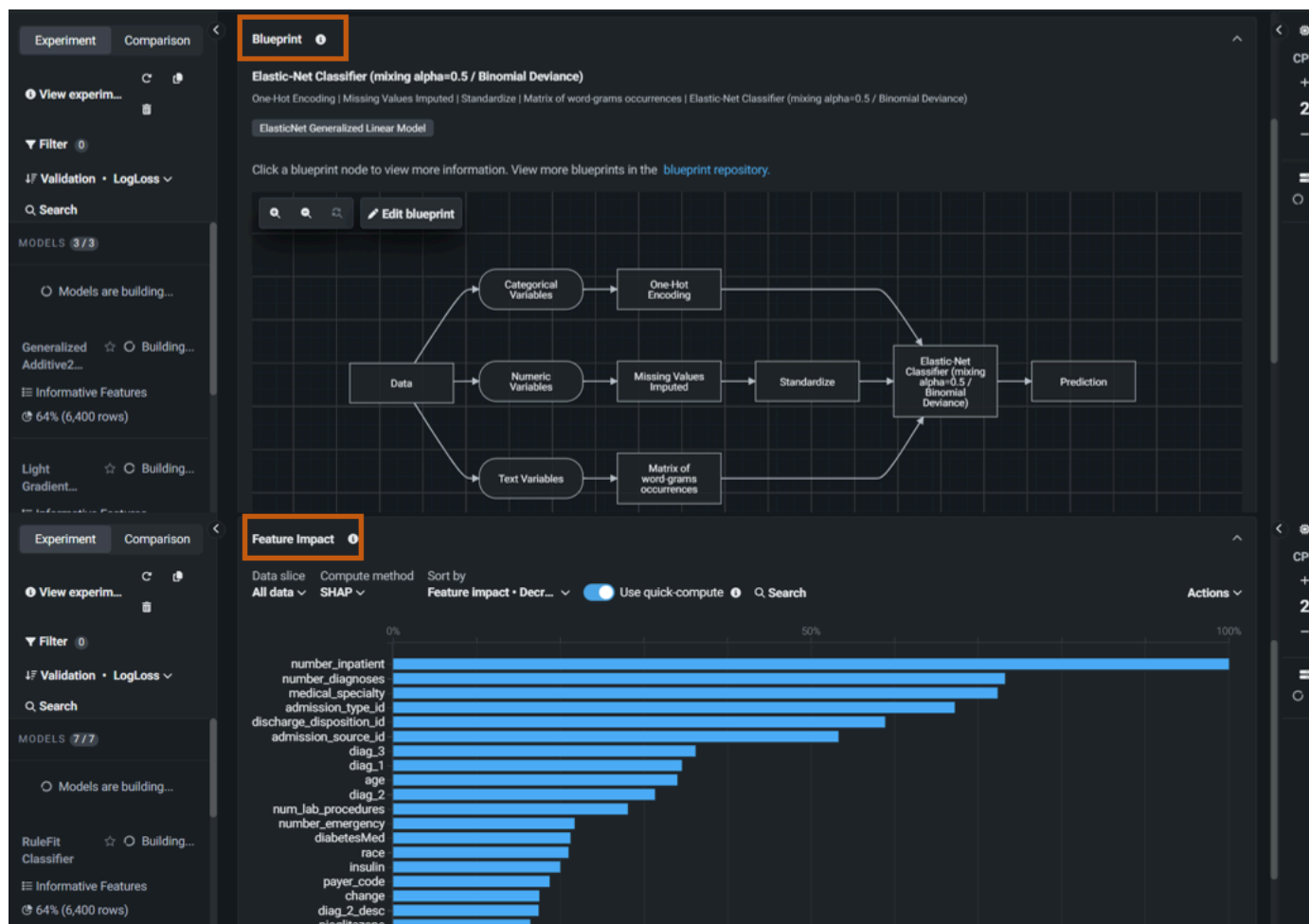
Step 18: Building models will take a while.

The screenshot shows the DataRobot Workbench interface. The top navigation bar includes the DataRobot logo, 'Workbench', and a breadcrumb trail: 'Workbench directory / Untitled_4 / < HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57'. On the right, there's a user profile 'trial-c3398e81-5e8' and a notification 'Days remaining: 29'. The left sidebar has tabs for 'Experiment' and 'Comparison'. Under 'Experiment', there are options to 'View experiment info', a 'Filter' (0), 'Validation' (LogLoss), and a 'Search' bar. Below this, a 'MODELS' section shows '0/0' models. The main area displays a large circular progress indicator and the text 'Models are building...'. The bottom of the sidebar shows details for two models: 'Keras Slim Residual Neural Network Classifier using...' and 'Elastic-Net Classifier (L2 / Binomial Deviance)', both with 'Informative Features' and '64% (6,400 rows)'.

Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

The screenshot shows the 'Model Overview' page in DataRobot Workbench. The top navigation bar is the same as the previous screenshot. The left sidebar now shows 'MODELS' with '3/3' models. The main area is titled 'Model Overview' and displays details for the 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)'. The 'Training scores: LogLoss' section shows: Validation (0.6089), Cross-validation (Score), and Holdout (0.6111). The 'Training settings' section shows: Training feature list, Informative Features (64% (6,400 rows)), and Training sample size. Below this, there are sections for 'Blueprint', 'Feature Impact', 'Feature Effects', 'Individual Prediction Explanations', and 'ROC Curve'. The left sidebar shows the 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)' model selected, with a score of 0.6089. Other models listed are 'Elastic-Net Classifier (L2 / Binomial Deviance)' with a score of 0.6115 and 'Keras Slim Residual Neural' with a score of 0.6327.

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.



Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.

The screenshot displays the Elastic-Net Classifier interface, specifically the "Model Overview" and "Make Predictions" sections.

Model Overview:

- Training scores:** LogLoss (Validation: 0.6089, Cross-validation: 0.6111, Holdout: 0.6111).
- Training settings:** Training feature list, Training sample size, Informative Features (64% (6,400 rows)).

Make Predictions:

- Make new predictions:** Prediction dataset (Prediction datasets cannot exceed 5.00 GB).
- Prediction options:**
 - ☒ Include additional feature values in prediction results. When enabled, results include specified feature values taken from the prediction dataset.
 - ☒ Add all features ☐ Add specified features
 - All features selected
 - ☐ Include Prediction Explanations
- Download recent predictions:** Predictions are stored and available to download for 48 hours. No recent predictions.

Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

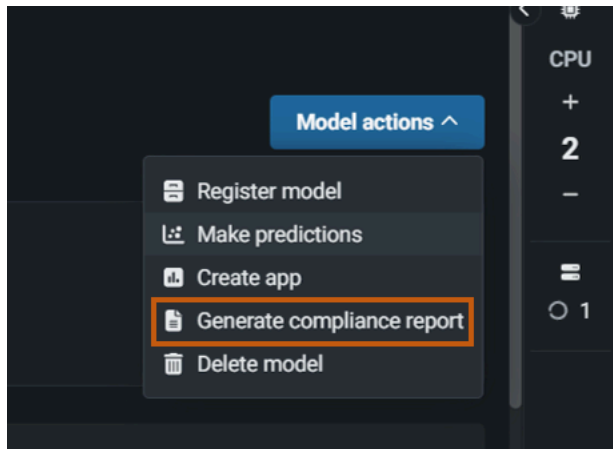


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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

[Dr. Pooja](#)



Skills Network